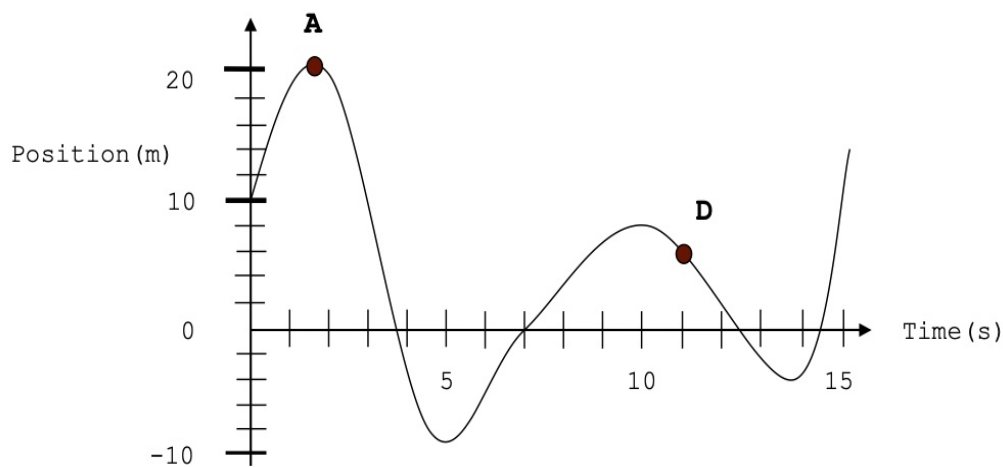
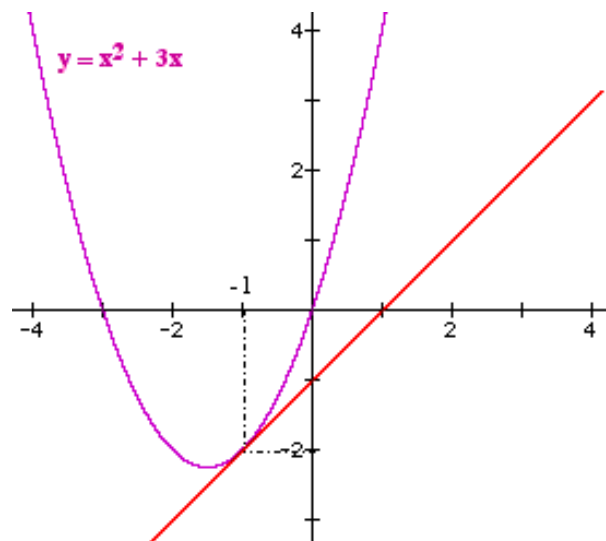


Assignment-02

- Consider the function $f(x) = 2 - x^2$; $x_0 = -1$ and $x_1 = 2$.
 - Find the instantaneous rate change of function at x_0 .
 - Find the average rate of change of function in the interval $[x_0, x_1]$.
 - Find the equation of the tangent line and secant line along with the graph of $f(x)$.
 - Explain your findings in (c).
- The following figure represents a position function of a particle at time t .
 - Find the average velocity over the time from A to D.
 - Find the value(s) of t at which the instantaneous velocity is zero.
 - Roughly sketch the velocity graph of the particle.



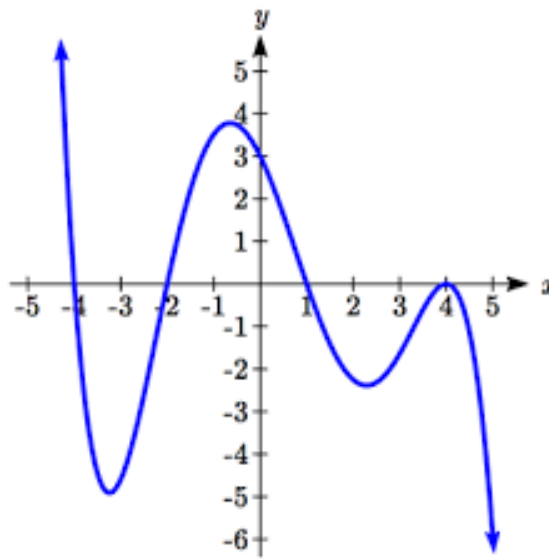
- Using the definition find the slope of the following curve at the prescribed point drawn below and hence find the equation of the given line.



- Using definition find the derivative with respect to x of $f(x) = \frac{1}{2x}$, and use it to find the equation of the tangent line to $f(x)$ at $x = 3$.

Assignment-02

5. Check the continuity and differentiability of the function $f(x) = \begin{cases} x^2 - 4 & ; x \leq 2 \\ -x + 2 & ; x > 2 \end{cases}$ at $x = 2$. Also sketch the graph of function.
6. Test the continuity and differentiability of $f(x) = \begin{cases} -x & ; x < 0 \\ x^2 & ; x \geq 0 \end{cases}$ at $x = 0$? Justify your answer graphically.
7. Sketch the graph of $f(x) = \begin{cases} x^2 - 1 & , x \leq 2 \\ 2x - 1 & , x > 2 \end{cases}$. Determine whether the function $f(x)$ is continuous and differentiable at $x = 2$.
8. Sketch the derivative graph of the function given in the following figure.



9. The following table defines the values of $f(x)$, $g(x)$, $f'(x)$ and $g'(x)$ at x . For $u(x) = f(x)g(x)$, and $v(x) = \frac{f(x)}{g(x)}$, find $u'(-1)$, and $v'(0)$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
-1	4	-2	7	-3
0	7	3	-1	-2

10. Use chain rule to evaluate the following derivatives:

- (a). If $x = \tan u$ and $u = t^3 - 2t \cos t + 5$, then find $\frac{dx}{dt}$
- (b). If $y = \cot^3 \sqrt{2 - 3 \sin x}$, then find $\frac{dy}{dx}$
- (c). Find $\frac{dy}{dx}$ of the functions $y = \sqrt{x^4 + \cos 3x}$ and $y = x^3 \sec \frac{1}{x}$
- (d). Find $\frac{dw}{dt}$, if $w = \sin 2x$ and $x = \sqrt{1 + \cos t^2}$
- (e). Find $\frac{dw}{dt}$, where $w = x^2$, $x = \sin y$, $y = \sqrt{t}$
- (f). Find $\frac{dy}{dx}$ for $y = \frac{(\ln(\tan x))^2}{x^3}$